

Real or Fake? Transformation Robustness Does Not Guarantee Cross-Generator Generalization

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Abstract. I started with a practical question: could a small classifier combine semantic and frequency features well enough to survive the image changes common on social platforms? The first detector joined a frozen 512-dimensional OpenCLIP embedding with a fixed 32-dimensional radial Fourier descriptor and fitted a logistic-regression head on CIFAKE. It reached 0.988409 clean ROC AUC, 0.956211 mean AUC across 14 transformation conditions, and a 0.972310 composite score. The ablation showed that augmentation, rather than the Fourier branch by itself, provided most of the robustness gain.

The blind test changed the direction of the work. On balanced SID-Set, COCO/DALL-E, and LAION/DALL-E samples, the selected hybrid produced AUCs of 0.4975, 0.5025, and 0.5000 and called every image fake. Its semantic-only component retained AUCs of 0.8870, 0.7680, and 0.7406. The largest absolute standardized frequency feature was 12.15 on clean CIFAKE but about 498–508 on the blind sets. Those values drove logits as high as 121.44 and saturated the sigmoid output.

I removed the frequency branch and retrained a semantic-only probe on balanced CIFAKE, SID-Set, and WildFake data. The resulting model uses 48,000 training rows and passes the recorded gates with 0.991900 SID validation AUC, 0.902925 COCO/DALL-E AUC, and 0.912700 LAION/DALL-E AUC. Its CIFAKE clean and robust AUCs fall to 0.957820 and 0.896636. The main result is straightforward: robustness to post-processing and transfer across generators are different objectives. A promotion process needs to test both, along with score saturation and branch-level feature ranges.

Keywords: AI-generated image detection; transformation robustness; domain generalization; OpenCLIP; frequency analysis; synthetic media forensics.

Artifact status. This project report records two distinct stages: the original hybrid candidate and the semantic-only model promoted after blind-test diagnosis. Their metrics and SHA-256 hashes remain separate. This is not a peer-reviewed publication.

Key finding. The 0.972310 CIFAKE composite and 0.5000 blind AUC are both correct. They answer different questions about the same frozen hybrid model.

1 Introduction

An image detector can fail for at least two different reasons. The first is post-processing: JPEG recompression, blur, resizing, noise, colour adjustment, or cropping. The second is distribution shift caused by a new generator, image source, resolution, compression history, or content domain. Both can reduce performance, but success on one does not establish success on the other.

TikTok TechJam 2026 Track 5 asks for clean and transformed performance, a continuous confidence score, reproducible evaluation, and an honest discussion of generalisation and false positives [1]. The workshop also suggests combining semantic and frequency evidence. I treated that combination as a starting hypothesis, not as a conclusion.

I organised the work around four questions. **RQ1** asks whether deterministic augmentation preserves ranking across the 14 required transformation conditions. **RQ2** asks what the semantic branch, frequency branch, and augmentation each contribute. **RQ3** asks whether the model selected on CIFAKE transfers to native-resolution images from unseen sources and generators. **RQ4** asks whether the failure diagnosis can produce a safer successor and promotion process.

The order of the experiments matters. I first built and selected the hybrid on CIFAKE, then froze that artifact before sampling the blind sets. The blind failure led to a branch-level investigation and a semantic-only, multi-domain training run. Reporting only the first score would hide the main lesson; reporting only the final model would hide how I arrived at it.

2 Related Work

Wang et al. showed that blur and JPEG augmentation can improve transfer for GAN-image detectors [2]. This supports the transformation protocol I used, but it does not establish robustness to every future generator.

Frank et al. found strong frequency artifacts in GAN-generated images and connected them to upsampling [3]. That makes frequency features a reasonable forensic hypothesis, but also a hypothesis that needs to be tested across newer generators and source pipelines.

Ojha et al. report that simple classifiers on fixed vision-language features can generalise better than networks trained directly on one real/fake split [4]. That result motivated my frozen-CLIP baseline and later supported the semantic-only remediation. PatchCraft uses texture-patch statistics and evaluates across 17 generator types [5]; it is a useful reminder that generator-held-out evaluation should be part of the design. Poletti et al. study a related transparency problem by comparing detector explanations with human judgments [6]. I did not add explainability to this detector, but their work reinforces the need to state clearly what a model score does and does not explain.

CLIP was trained to associate images and text across broad internet data and is often useful as a transferable representation [8]. OpenAI's account also describes the gap between benchmark performance and real-world stress tests [10]. That warning is central here: a broad representation does not prevent a narrow decision rule.

All figures in this report are original. They are generated from

the repository’s architecture or measured results; no third-party diagram is reproduced.

3 Model Design and Iteration

3.1 Task Definition

Given an RGB image x , the detector returns a score $p(x) \in [0, 1]$. Larger values rank the image as more likely to be generated. I use $y = 0$ for real and $y = 1$ for generated images. ROC AUC is the primary metric because it evaluates ranking over every threshold; accuracy and F1 are secondary diagnostics at a selected threshold.

3.2 Original Hybrid

Figure 1 shows the submitted design. Each image follows two fixed feature paths. I concatenate and standardize their outputs before passing them to logistic regression. The 151,277,313 parameters of the OpenCLIP backbone remain frozen.

The semantic branch uses OpenCLIP ViT-B-32-quickgelu with the openai checkpoint. If $f_{\text{CLIP}}(x) \in \mathbb{R}^{512}$ is the image encoder, the stored feature is

$$s(x) = \frac{f_{\text{CLIP}}(x)}{\|f_{\text{CLIP}}(x)\|_2}. \quad (1)$$

Features are extracted under half-precision autocast, normalized in floating point, cached as float16, and promoted to float32 before fitting.

The frequency path converts the image to grayscale, resizes it to 256×256 , subtracts the mean, and applies a two-dimensional Hann window before computing

$$M(u, v) = \log(1 + |\mathcal{F}\{x_w\}(u, v)|). \quad (2)$$

After shifting zero frequency to the centre, I divide normalized radius into 32 equal-width bins:

$$r_k(x) = \frac{1}{|B_k|} \sum_{(u,v) \in B_k} M(u, v), \quad k = 1, \dots, 32. \quad (3)$$

The descriptor has no learned parameters.

For the hybrid, $z(x) = [s(x); r(x)] \in \mathbb{R}^{544}$. I fit a StandardScaler on the training features, followed by L2-regularized logistic regression [9]:

$$p(x) = \sigma(w^\top \text{std}(z(x)) + b). \quad (4)$$

The interface calls this value $P(\text{FAKE})$, but it is not calibrated to deployment prevalence. It is a ranking score, not proof of a literal real-world probability.

3.3 Promoted Semantic Model

The successor removes $r(x)$ and uses the same scaler-plus-logistic-regression pattern on the 512-D semantic feature. I balanced its 24,000 source images across CIFAKE, SID-Set, and WildFake, with 4,000 real and 4,000 generated images per

Table 1. Individual transformation protocol.

Family	Values and implementation
JPEG	Quality 90, 70, 50, 30; encode/decode, 4:2:0
Blur	$\sigma = 0.5, 1.0, 2.0$; Gaussian blur
Resize	0.5, 0.25; bicubic downsample then restore
Noise	$\sigma = 0.02, 0.05, 0.10$; additive RGB noise
Jitter	$\pm 20\%$ brightness, contrast, saturation
Crop	Centre 80% retained, then bicubic restore

domain. Each image contributes one clean view and one deterministically transformed view, producing 48,000 fitting rows.

For WildFake fitting, I used nine allowed groups: ADM, DDPM, GALIP, GigaGAN, VQGAN, VQVAE, AFHQ, CelebA-HQ, and LSUN-Church. COCO val2017, DALL-E Advanced, and all frozen evaluation identifiers are excluded. The threshold, 0.7819586396, was selected on a disjoint 500-real/500-generated SID calibration split. The final head has 512 coefficients and one intercept.

4 Data and Evaluation Protocol

4.1 CIFAKE and Transformations

CIFAKE contains 32×32 JPEG images derived from CIFAR-10 real images and generated counterparts [7]. The published split is preserved: 50,000 real and 50,000 generated training images, followed by 10,000 real and 10,000 generated test images. Test labels are not used for fitting. All 20,000 test images are scored under every condition.

I apply transformations individually rather than stacking them. For augmented fitting, family and severity are selected deterministically from the SHA-256 of the global seed, image identifier, and transform key. Each image appears once in its clean form and once in a transformed form.

If A_0 is clean AUC and A_j is AUC under transformed condition j , then

$$A_{\text{robust}} = \frac{1}{14} \sum_{j=1}^{14} A_j, \quad S = 0.5A_0 + 0.5A_{\text{robust}}. \quad (5)$$

For a secondary family-balanced score, I first average within each family and then across the six families.

4.2 Frozen Blind Samples

I froze the original hybrid artifact before selecting three balanced 400-image samples. Each sample contains 200 real and 200 generated images. The manifests preserve identifiers, paths, labels, dimensions, byte sizes, and hashes.

SID validation uses label 0 and label 1 images from SID-Set revision dc03ead... [11]. WildFake default uses COCO val2017 and DALL-E Advanced; a matched configuration uses LAION-5B and DALL-E Advanced [12]. Sampling seeds are 2028, 2026, and 2027. The frozen hybrid SHA-256 is e9bc59e4...c3427b5e. Five thousand stratified bootstrap replicates estimate 95% AUC intervals.

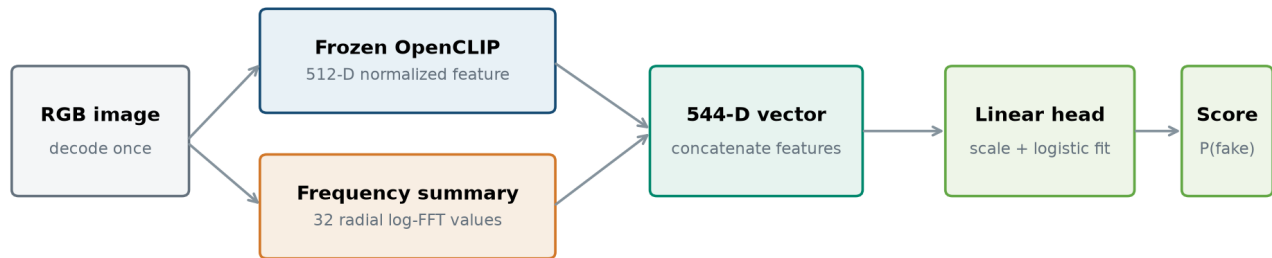


Figure 1. Feature pipeline for the submitted hybrid detector. The frozen 512-D semantic branch and fixed 32-D radial-frequency branch form a 544-D representation. The standardization statistics and logistic head are learned; the backbone and frequency descriptor are fixed.

4.3 Promotion Gates

After the diagnosis, I made these held-out source configurations explicit promotion gates for the semantic-only successor. Fitting excludes their identifiers, but I used the gates to decide whether to promote the model. They are strong held-out validation evidence, not a new untouched post-promotion blind test. A genuinely independent evaluation remains future work. The promoted SHA-256 is `0c1cf7d6...9208449cf`.

4.4 Compute

I ran the experiments on Windows 11 with an AMD Ryzen 7 7800X3D, 32 GB RAM, and Radeon RX 7900 XTX with 24 GB VRAM. The validated stack uses Python 3.12.13, PyTorch 2.9.1 with ROCm 7.2.1, torchvision 0.24.1, OpenCLIP 3.3.0, NumPy, Pillow, and scikit-learn. The final native feature extraction, fit, gate evaluation, and promotion pipeline completed in 547.063 seconds under a hard 3,600-second budget.

5 What Worked on CIFAKE

5.1 Transformation Robustness

The original hybrid performed well across most of the challenge matrix. JPEG remained above 0.96 AUC even at quality 30. Mild blur, moderate resizing, colour jitter, and cropping retained useful ranking. Quarter-scale resizing and blur at $\sigma = 2.0$ were the first conditions below 0.90.

Figure 2 shows useful robustness to ordinary post-processing until an operation removes a large amount of image information. It does not, however, test a new generator or source pipeline.

5.2 What Supplied the Gain

The ablation in Table 2 and Figure 3 separates semantic features, frequency fusion, and augmentation. Adding FFT features raises clean AUC by 0.003005 but lowers robust AUC by 0.006829. Adding augmentation then lifts robust AUC by 0.045270 relative to the clean hybrid.

The frequency branch alone did not create the robustness gain. The improvement appeared after the hybrid saw deterministic transformations during training.

5.3 Benchmark Trade-Off

The promoted semantic model is not a free improvement: it gives up some CIFAKE performance in exchange for substantially better native-resolution transfer.

Table 3. CIFAKE trade-off between artifact stages.

Artifact stage	Clean	Robust	Composite
Original hybrid	.988409	.956211	.972310
Promoted semantic	.957820	.896636	.927228

For the promoted model, the hardest CIFAKE conditions are blur $\sigma = 2.0$ at 0.785540 AUC, resize 0.25 at 0.774426, and noise $\sigma = 0.10$ at 0.797451. This trade-off is important and should not be hidden behind the stronger cross-generator numbers.

6 Where the Hybrid Failed

6.1 Blind Transfer Reversed the Ranking

The blind result is the turning point of the project because it overturns the benchmark ranking. The semantic baseline retains ranking information on all three samples. Both hybrid variants are near chance, and the selected augmented hybrid calls every image fake at threshold 0.5.

The default WildFake sample has an obvious image-size association, but the failure remains in the LAION-matched configuration, where size-only AUC is 0.5. Resolution mismatch alone therefore does not explain the collapse.

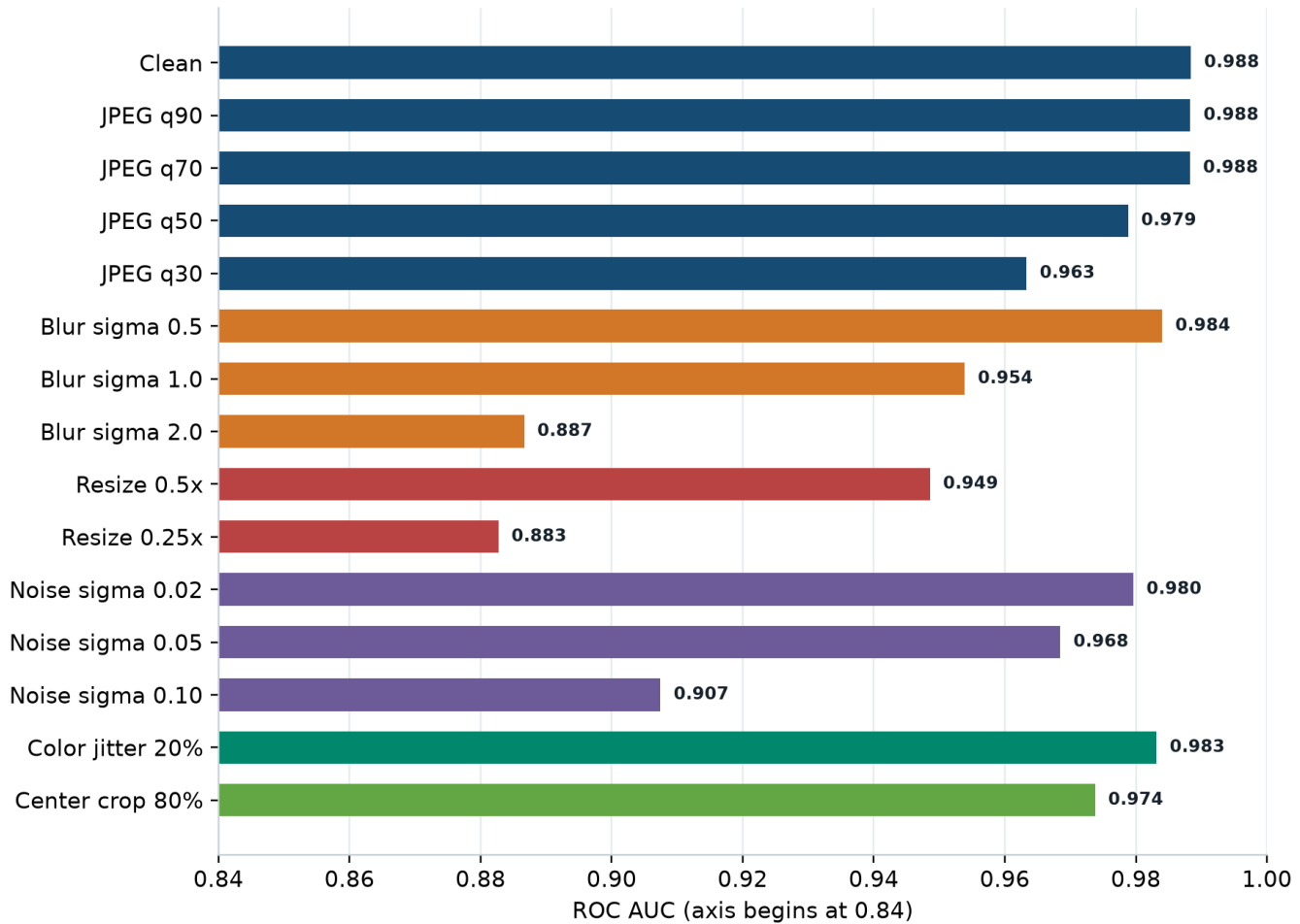


Figure 2. ROC AUC for the original hybrid under individual transformations. Every bar uses the same 20,000 held-out CIFAKE images. The x-axis begins at 0.84; labels report absolute AUC.

Table 2. Original CIFAKE ablation. Robust AUC is condition-weighted over 14 transforms.

Model and fitting data	Clean AUC	Robust AUC	Composite
Semantic, clean	0.988755	0.917770	0.953263
Semantic + FFT, clean	0.991760	0.910941	0.951351
Semantic + FFT, clean + augmented	0.988409	0.956211	0.972310

6.2 Frequency Extrapolation

The branch-level values provide a concrete mechanism. I fitted the original frequency standardization recipe on 200,000 clean-plus-augmented training rows. The training 99.9th percentile of absolute standardized frequency values is 5.48, the training maximum is 15.34, and the clean CIFAKE test maximum is 12.15.

The blind maxima are 507.99 on SID-Set, 498.14 on COCO/DALL-E, and 505.87 on LAION/DALL-E. On SID, total hybrid logits range from 7.96 to 121.44. Since $\sigma(7.96) \approx 0.99965$, even the minimum is already close to one.

These values explain why the threshold metrics collapse and why changing the threshold after the fact would not restore ranking. The semantic branch is not well calibrated by itself: its median real score on SID is still 0.986973. It does retain order information, but the frequency branch erases that order by

dominating the logit outside the fitting distribution.

7 Remediation and Promotion

Before replacing `outputs/model.joblib`, I required the semantic-only successor to pass every recorded gate.

Table 5. Promoted semantic-only gates. PFR uses threshold 0.7819586396.

Gate	ROC AUC	PFR	Result
SID calibration	0.993972	0.5130	Pass
SID validation	0.991900	0.5075	Pass
COCO / DALL-E	0.902925	0.5275	Pass
LAION / DALL-E	0.912700	0.4050	Pass

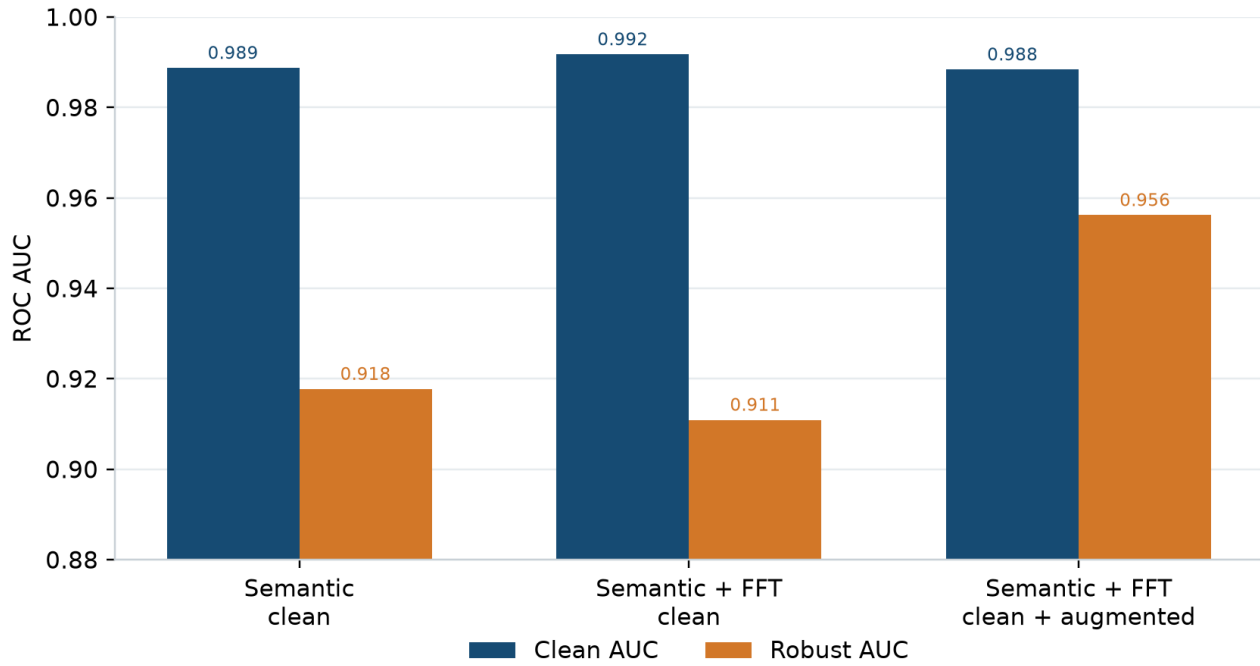


Figure 3. Clean and robust ROC AUC for three CIFAKE feature and training configurations. Robust AUC is the condition-weighted mean over 14 transformations.

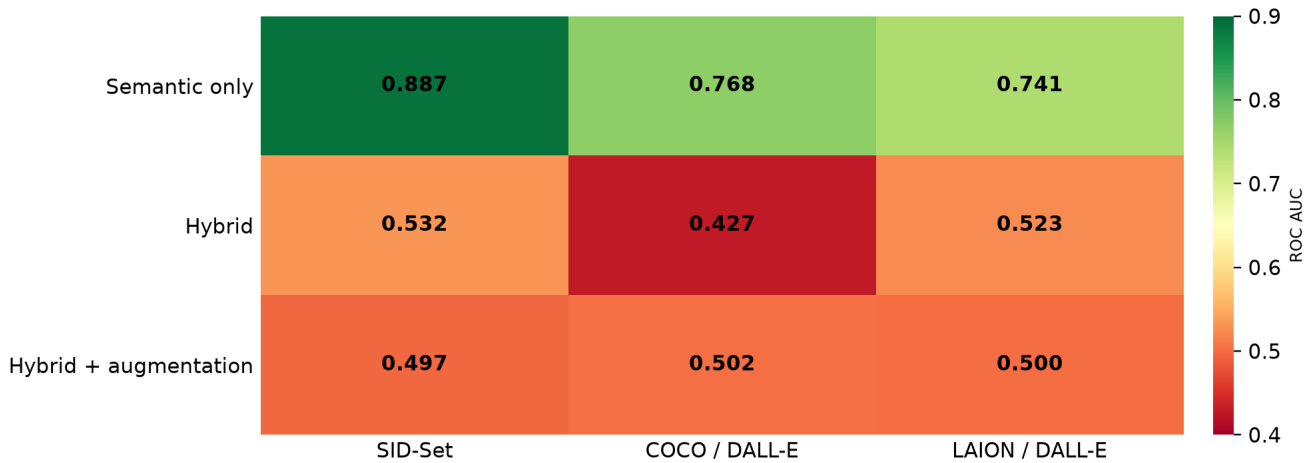


Figure 4. ROC AUC on three frozen, balanced native-resolution samples for the three model variants. Each cell uses the same 200-real/200-generated sample.

The successor removes the observed saturation and improves held-out ranking. That does not prove universal generalisation. Because these datasets influenced promotion, the next credible test must use generators and sources that were not used for fitting, calibration, or model selection.

Figure 6 summarises the iteration. The practical conclusions are clear. Augmentation is local: training on known transformations teaches useful invariance around CIFAKE, not automatic invariance to a different generator or source. Plausible features can become shortcuts; unconstrained linear contributions are dangerous when standardized values leave the fitting range. Finally, model selection defines what “best” means. The CIFAKE composite selected the right model for that matrix and the wrong model for broad transfer.

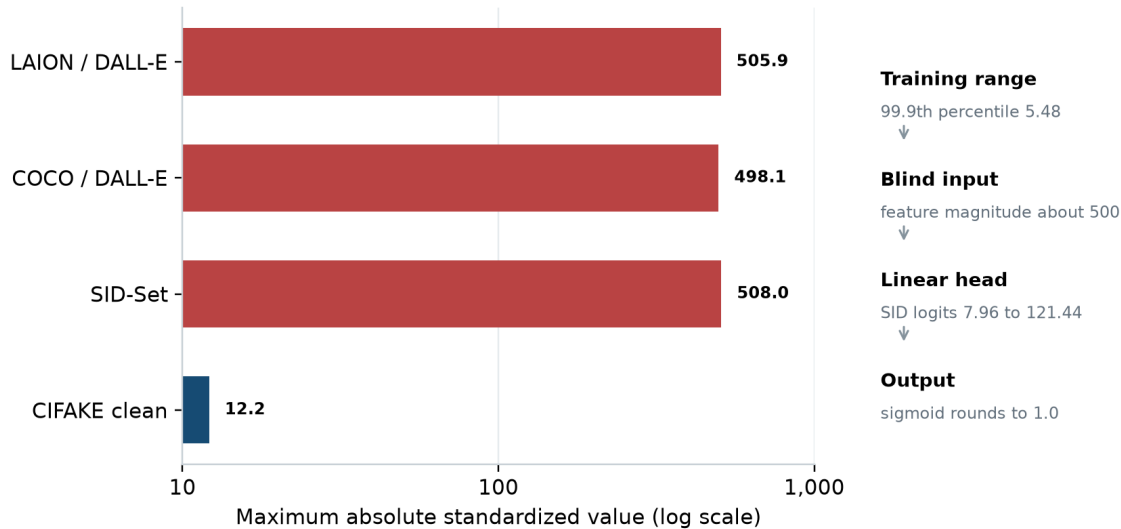
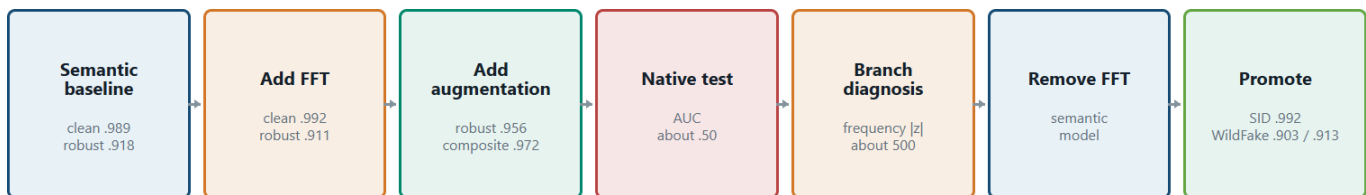
8 Limitations and Responsible Use

CIFAKE is low-resolution and generator-narrow; its transformed test set still shares the same underlying dataset. The original blind samples and current gates contain only 400 images each. I used the native-resolution gates to select the promoted model, so they are no longer an untouched final test. WildFake fitting covers nine groups, but many generators, editing systems, styles, and capture pipelines remain unseen.

I evaluated transformations individually, while platform pipelines may stack resizing, recompression, colour adjustment, and sharpening. Scores are not calibrated to deployment prevalence. Demographic bias, adversarial attacks, partial edits, mixed real/generated images, and fairness were not evaluated.

Table 4. Blind ROC AUC by frozen model variant. Each cell uses the same 200-real/200-generated sample.

Frozen model	SID-Set	COCO / DALL-E	LAION / DALL-E
Semantic	0.886975	0.767975	0.740600
Hybrid	0.532475	0.427500	0.523488
Hybrid + augmentation	0.497500	0.502500	0.500000

**Figure 5.** Frequency-branch diagnosis. The left panel compares the maximum absolute standardized value on clean and blind inputs; the right panel lists the resulting range shift through the linear head.**Figure 6.** Model variants and evaluation steps used during development, from the semantic baseline through hybrid training, native-resolution evaluation, branch diagnosis, and semantic-only promotion.

OpenCLIP may inherit biases from pretraining.

This detector is a research prototype. It should not be the sole basis for content removal, account suspension, copyright decisions, fraud allegations, academic penalties, or authorship claims. False positives can harm legitimate creators; false negatives can create false confidence. Human review, provenance metadata, source verification, and multiple independent signals remain necessary.

Interpretation. The promotion gates are strong held-out evidence, but not a fresh post-promotion blind test. The next version needs an independent generator-and-source evaluation.

produces deterministic records containing `image_path` and continuous `pred`. The artifact validates its schema and feature configuration and rejects non-finite predictions.

I froze both stages, their hashes, plotted values, and provenance in the report snapshot `docs/report_metrics.json`. This prevents later promotion runs from silently rewriting historical figures. I rebuild all six figures and the PDF with:

```
powershell -ExecutionPolicy Bypass -File
scripts/build_technical_report.ps1
```

The figure script uses Pillow and repository data rather than downloaded illustrations.

9 Reproducibility

The judge-facing prediction contract for my CLI is:

```
python src/predict.py -input_dir images -out
preds.json -device auto
```

The CLI recursively finds JPEG, PNG, and WebP inputs and

10 Conclusion

The original hybrid solved the problem I selected it to solve. Deterministic augmentation raised its mean transformed CIFAKE AUC to 0.956211 and composite to 0.972310. The frequency branch alone did not produce that gain.

Blind testing then showed that the selected model was not a general detector. It ranked three native-resolution samples at chance and saturated every output as fake. A measured forty-fold jump in standardized frequency magnitude explained why.

Removing that branch and fitting a semantic-only probe on balanced, multi-domain data produced a better promotion candidate. The candidate passed the recorded SID and WildFake gates, although its CIFAKE composite fell to 0.927228. This is an engineering trade-off, not a contradiction.

My next step is an independent post-promotion evaluation on generators and source pipelines that did not influence fitting, calibration, or gating. Until then, the strongest supported claim is narrow but useful: transformation robustness and cross-generator transfer must be measured separately, and branch-level out-of-distribution checks can prevent a strong benchmark score from promoting a brittle model.

References

- [1] TikTok TechJam 2026. Robust detection of AI-generated images under real-world transformations. Track 5 problem statement and technical workshop notes, 2026.
- [2] S.-Y. Wang, O. Wang, R. Zhang, A. Owens, and A. A. Efros. CNN-generated images are surprisingly easy to spot... for now. In *Proceedings of CVPR*, 2020. <https://arxiv.org/abs/1912.11035>.
- [3] J. Frank, T. Eisenhofer, L. Schönherr, A. Fischer, D. Kolossa, and T. Holz. Leveraging frequency analysis for deep fake image recognition. In *Proceedings of ICML*, 2020. <https://arxiv.org/abs/2003.08685>.
- [4] U. Ojha, Y. Li, and Y. J. Lee. Towards universal fake image detectors that generalize across generative models. In *Proceedings of CVPR*, 2023. <https://arxiv.org/abs/2302.10174>.
- [5] N. Zhong, Y. Xu, S. Li, Z. Qian, and X. Zhang. PatchCraft: Exploring texture patch for efficient AI-generated image detection. arXiv:2311.12397v3, 2024. <https://arxiv.org/abs/2311.12397>.
- [6] S. Poletti, J. Ilyes, M. Hasenbalg, D. Fischinger, and M. Boyer. AI-generated images: What humans and machines see when they look at the same image. arXiv:2605.06143, 2026. <https://arxiv.org/abs/2605.06143>.
- [7] J. J. Bird and A. Lotfi. CIFAKE: Image classification and explainable identification of AI-generated synthetic images. *IEEE Access*, 12:15642–15650, 2024. <https://doi.org/10.1109/ACCESS.2024.3356122>.
- [8] A. Radford et al. Learning transferable visual models from natural language supervision. In *Proceedings of ICML*, 2021. <https://arxiv.org/abs/2103.00020>.
- [9] F. Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [10] OpenAI. CLIP: Connecting text and images. 5 January 2021. <https://openai.com/index/clip/>.
- [11] SID-Set. Hugging Face dataset repository, revision dc03ead57929879319ce30a82bfcfb8d317b10bd. https://huggingface.co/datasets/saberz1/SID_Set.
- [12] WildFake. ModelScope dataset repository and TechJam evaluation subset. <https://modelscope.cn/datasets/hy2628982280/WildFake/summary>.

A Claim-to-Artifact Map

Original benchmark metrics. `outputs/clean_metrics.json`; frozen historical conditions in `docs/report_metrics.json`.

Frozen blind evaluation. `outputs/cross_domain_summary.json`; branch diagnostics in `data/blind-test/ablation_results.json`.

Promoted model. Training recipe and gates in `outputs/native_metrics.json`; per-condition results in `outputs/robustness_table.csv`; wall-clock stages in `outputs/native_training_timing.json`.

Inference and report provenance. `src/predict.py`; immutable plotted values and recipes in `docs/report_metrics.json`.